Parbhani Drought Monitoring Project Full Workflow Report & Output Reference

Region: Parbhani District, Maharashtra, India Study period: 2016-01-01 2026-10-06 (130 months) Baseline: 1991-2020

Generated: 2026-10-07 Pipeline: exit 0 Smoke API: 9/9 Map QC: ALL PASS (7/7)

Region	Parbhani District 9 tehsils (Gangakhed, Jintur, Manwath, Palam, Parbhani, Pathri, Purna, Selu, Sonpeth)
Area	6,307.49 km (tehsil sum = district polygon, diff 0.000%)
Study period	2016-01-01 2026-10-06 (130 months)
Baseline (SPI Gamma fit)	1991-01-01 2020-12-31
CRS	EPSG:32643 (UTM 43N) for all rasters and zonal work
Total outputs	564 files, 0.79 GB
Runtime	~32 s (cached rasters)
CDI formula	0.35 SPI-3 + 0.25 SSMI + 0.25 VCI_z + 0.10 EVI_z + 0.05 PCI_z
Stage thresholds	0 No Drought (1, 0) Mild (1.5, 1] Moderate (2, 1.5] Severe 2 Extreme

1. The Workflow (9 stages in src/main.py)

1 shapefile load 9-tehsil boundary, validate CRS / areas
2 gee_auth non-interactive auth (project dark-influence-491307-n3)
3 weather Open-Meteo archive -> precip / ET0 / temp -> SPI-1/3/6/12
(Gamma distribution fit on 1991-01-01 - 2020-12-31)
4 gee_sentinel2 2,293 scenes -> 129 monthly composites -> NDVI / EVI / PCI
rasters at 100 m
5 soil_moisture NASA SMAP L4 (11 km) monthly means -> per-tehsil z-score = SSMI
6 zonal zonal statistics per tehsil x month from every raster
7 classify per-pixel CDI -> DROUGHT_STAGE_<yyyy-mm>.tif -> tehsil stages
8 sampling 4,498 stratified points -> Random Forest (2 label sets)
9 viz 8 figures + 9 maps + 9 map PDFs + contact sheet + 2 dashboards
verify per-stage checks -> outputs/reports/summary.md

2. Outputs outputs/rasters/ (517 files, 0.73 GB)

The spatial core of the project: every downstream table and map is derived from these GeoTIFFs.

File pattern	Count	Meaning
S2_NDVI_YYYY-MM.tif	129	Monthly Sentinel-2 NDVI composite, 100 m, float. 2016-02 absent (no usable overpass).
S2_EVI_YYYY-MM.tif	129	Enhanced Vegetation Index, same grid.
S2_PCI_YYYY-MM.tif	129	Precipitation Condition Index (vegetation response to rainfall).
classified/DROUGHT_STAGE_YYYY-MM.tif	130	FINAL PRODUCT per-pixel CDI mapped to stage class 0 4, uint8, EPSG:32643. One per month including 2016-02 (weather/SMAP only).
stack/	0	Reserved (empty).

387 Sentinel-2 rasters + 130 classified = 517. Cloud handling: scene filter <60% cloud (fallback <80%) plus pixel-level Cloud Score+ (cs_cdf 0.6). SR tier used for 119 months, TOA fallback for 10.

3. Outputs outputs/csv/ (13 tables)

File	Rows	What it holds
weather_spi.csv	430	Full monthly weather (precipitation, ET0, water deficit, temperature) + SPI-1/3/6/12 + drought class labels, 1991 2026. The climate backbone.
tehsil_zonal_indices.csv	1,170	9 tehsils 130 months: weather + SSMI + NDVI/EVI/PCI zonal means the pre-CDI table.
zonal_ndvi/evi/pci.csv	1,161 each	Raw zonal statistics (mean, median, std, min, max, valid_pixel_count). 1,161 = 9 129 because 2016-02 has no Sentinel-2 raster.
soil_moisture_ssmi.csv	1,170	SMAP sm_mean per tehsil + z-scored SSMI.
tehsil_drought_stages.csv	1,170	MASTER RESULTS TABLE every component (SPI, SSMI, VCI, EVI_z, PCI_z and all z-scores), CDI, stage, stage_name, dashboard_class, tehsil_area_km2. Everything downstream reads this.
stage_area_summary.csv	5,850	tehsil month stage area_km2, pct_area. Feeds the stage maps' area tables and legends.
sentinel2_indices.csv	129	Per-month composite provenance: scene count, tier (SR/TOA), mask method, district mean NDVI/EVI/PCI.
point_index_values.csv	580,242	Every sample point month with NDVI/EVI/PCI/SPI-3/SPI-6/SSMI the Random Forest feature matrix.
stratified_sample_points.csv	4,498	Point IDs, UTM + lon/lat, stratified by tehsil.
rf_classification_report.csv	6	RF metrics with COMPOSITE CDI STAGE labels accuracy 77.6%, weighted F1 0.788.
rf_classification_report_spi3_stage.csv	7	RF metrics with SPI-3-ONLY STAGE labels accuracy 90.7%, weighted F1 0.905.

Stage distribution (1,170 tehsil-months)

Stage	Class	Count	CDI range observed
0	No Drought	674	0.00 +1.95
1	Mild Drought	406	0.99 0.00
2	Moderate Drought	69	1.50 1.00
3	Severe Drought	21	1.89 1.50
4	Extreme Drought	0	(not reached in study period)

Observed CDI ranges match the configured thresholds exactly.

4. Model & reports

outputs/models/

drought_rf_model.joblib trained Random Forest. Features are **only** SPI / SSMI / NDVI / EVI / PCI (CDI is never fed back as a feature). Two label variants are evaluated in the two report CSVs.

outputs/reports/

summary.md / summary.json execution report: stage timings, row counts, mask/tier strategy, data-availability gaps, assumptions, library versions, and the full file tree with sizes.

drought_dashboard_data.json pre-aggregated payload served by the API */dashboard* endpoint.

workflow_report.pdf this document.

5. Figures outputs/figures/ (8 PNGs, 300 dpi)

Figure	What it shows / how to read it
SPI_NDVI_trend.png	Dual-axis time series 2016 2026. Left axis: SPI-3 (solid blue) and SPI-6 (dashed orange), dotted zero line, three dashed threshold rules (1 / 1.5 / 2), red shading where SPI-3 1 = drought periods. Right axis (green): Sentinel-2 mean NDVI the line BREAKS over cloud-masked months instead of interpolating. The incomplete last month (2026-10, data to Oct 6) is shaded with a labelled band. Legend sits outside the plot area below it.
SPI3_bar_drought_classification.png	Monthly SPI-3 bars colour-coded by drought class (green yellow orange red dark red) with zero line and threshold rules. Instant 'which months were dry' reading.
correlation_heatmap.png	Pearson correlation of precipitation, ET0, water deficit, temperature, SPI-1/3/6/12, SSMI and the Sentinel-2 indices. S2 months are joined to weather by year-month string (weather is month-END, S2 is month-START a naive date join would give all-NaN). Pairwise-complete observations per cell.
tehsil_boxplots.png	Per-tehsil distribution of the key indices and CDI shows how similar the 9 tehsils are (very similar: SPI is literally identical across them, see section 8).
drought_stage_heatmap.png	Tehsil (rows) month (columns) categorical heatmap of the assigned stage the at-a-glance drought history. Visually confirms that 99 of 130 months are a single stage district-wide.
index_distributions.png	2 3 histograms + KDE for SPI-3, SSMI, NDVI, EVI, PCI and CDI, with 1 and 2 reference lines on the drought indices distribution shape versus the thresholds.
rf_confusion_matrix.png	Hold-out confusion matrix for the Random Forest (composite CDI-stage labels). Matches the 77.6% accuracy report.
rf_feature_importance.png	Gini importance of the five features which driver dominates the prediction.

6. Maps outputs/maps/ (22 files)

Layout: 13 8.5 in at 300 dpi **3900 2550 px** PNG plus a vector PDF twin for every map. Header row (title + subtitle), map/panel row and footer row (data sources | CRS | prepared date) are separate GridSpec rows so text can never collide. Every map carries a north arrow, a scale bar and a Maharashtra locator inset.

6.1 Drought-stage maps map_stage_<yyyy-mm>.png/.pdf (4 2 = 8 files)

Map	Why chosen	Content
map_stage_2019-05	Driest month (district SPI-3 = 2.23)	All 9 tehsils Severe (stage 3), CDI 1.89 1.68
map_stage_2025-09	Wettest month	All No Drought (stage 0), CDI +1.62 +1.95
map_stage_2026-09	Latest complete month	All Mild (stage 1), CDI 0.60 0.42
map_stage_2026-10	Current month PROVISIONAL	Mixed Mild / Moderate, CDI 1.32 0.75, marked incomplete (data to 2026-10-06)

Each stage map contains: tehsils filled by stage colour **plus a hatch pattern** per stage (colour-blind safe, survives greyscale printing); tehsil name + stage + CDI labels inside the polygons; a **legend with all 5 classes** (absent stages greyed, shown as 0 km / 0.0%); a **stage-area bar** in km and % of district; a district-wide headline with SPI-driven subtitle; and a data table of stage, area and share of district.

6.2 Index maps map_index_<name>.png/.pdf (5 2 = 10 files)

Map	Source	Colour bar / extras
map_index_ndvi	Sentinel-2, 2016 2026 mean	YlGn scale; AREA-WEIGHTED district mean marked on the colour bar; hi/lo tehsils bolded in the table; caution note that tehsil spread is small; footer warns the stretched scale is not comparable between maps.
map_index_evi	Sentinel-2	Same design as NDVI.
map_index_pci	Sentinel-2	Same design as NDVI.

Map	Source	Colour bar / extras
map_index_spi3	Open-Meteo (single point)	Map is uniform by construction, so the right panel instead shows the full SPI-3 time series with threshold rules, the last month hatched, the drought rule labelled, plus a stats card (min/mean, % of months at 1 / 1.5 / 2, IMD/CHIRPS note).
map_index_ssmi	NASA SMAP L4	5-column table (No. | Tehsil | % months flagged | SSMI anomaly | Area km) plus a marker-based multi-panel; footer states the actual 11 km processing grid.

6.3 Supporting files

File	Purpose
contact_sheet.png	4-column thumbnail grid of all 9 maps for quick review.
data_audit.md	PHASE 2 AUDIT, RUN BEFORE DRAWING 10 checks: tehsil areas vs polygon (0.000% diff), stage-share totals (max deviation 0.000000%), simple vs area-weighted means (maps use area-weighted), CDI variance decomposition, SMAP pixel counts per tehsil, SPI sanity, partial-month status, last date per source, NaN inventory, footer emission. Warnings are reported, never hidden.
qc_report.md	PHASE 4 QUALITY GATE 7 checks, ALL PASS: (1) no text overlaps and nothing outside the figure, (2) every label inside its polygon or a numbered marker, (3) tables fit with no truncation and a clear footer, (4) area sum = district within 0.5% and stage shares = 100.0% and means area-weighted, (5) legends complete, colour bars unit-labelled, PNG = 3900 2550 px and PDF a real vector file for all 9 maps, (6) filenames match map_stage_<yyyy-mm> / map_index_<name> with PDF twins, (7) everything regenerated and the contact sheet fresh.
map_prompt.md	The specification the map code implements (Phases 1 4).

7. Dashboards & API

dashboards/

interactive_drought_dashboard.html Plotly dark-theme dashboard: SPI time series, stage distribution, NDVI/EVI, regional comparison and calendar-month SPI comparison, with hover, zoom and legend.

drought_stage_timeline.html Plotly tehsil month stage heatmap plus monthly stage-count bars: the interactive counterpart of drought_stage_heatmap.png.

API scripts/smoke_api.py 9/9 OK

Endpoint	Returns
GET /	Endpoint index.
GET /spi?tehsil=&start=&end=&window=	SPI-1/3/6/12 series.
GET /indices/{ndvi|evi|pci|ssmi}	Monthly index per tehsil.
GET /drought-stage?month= or ?tehsil=&start=&end=	CDI, stage, stage_name, dashboard_class, tehsil area.
GET /baseline?year=	1991 2020 monthly climatology.
GET /dashboard	Metadata + all tehsils + aggregates.

8. Verification & data issues (all reported, none hidden)

verify_outputs per stage

Stage	Result
shapefile	OK
zonal	OK
classification	OK
visualizations	OK requires 4 stage maps, 9 maps + 9 PDFs, data_audit.md and qc_report.md reporting ALL PASS

Stage	Result
sentinel2	3 known warnings (below)

Known, accepted gaps the only warnings in the whole run

2016-02 no usable Sentinel-2 scene, so no S2 rasters exist for that month.

2016-07 and 2024-07 composites fully cloud-masked, giving all-NaN NDVI/EVI/PCI rasters.

Structural findings documented in data_audit.md

SPI-3 is spatially constant one distinct value per month across all 9 tehsils, because weather comes from a single Open-Meteo point. All spatial CDI variation therefore comes from satellite and SMAP data: VCI 40.2%, EVI 25.5%, SSMI 21.4%, PCI 12.9%, SPI 0.0%.

99 of 130 months are a single stage district-wide CDI spread per month is 0.09 to 0.57 (mean 0.31), consistent with SPI's 0% share.

SSMI mean 0 by construction it is a per-tehsil calendar-month z-score.

Study-period SPI mean +0.30 against the 1991 2020 baseline FLAGGED, not corrected: data and formula were left untouched.

2026-10 is partial (6 of 31 days) so SPI may be biased dry; the map is labelled PROVISIONAL.

SMAP grid is 11 km (config scale = 11000), not the '~9 km' native product size reported explicitly with a per-tehsil pixel table (10 20 pixels each, 132 over the district).

Boundary area 6,307 km vs official ~6,500 km 2.96% difference, flagged WARN; the project boundary remains authoritative.

9. Folder tree (abridged)

```
D:\Drought_Monitouring Project\
|-- config\config.yaml              dates, CDI weights, thresholds, SMAP settings
|-- data\Parbhani\Parbhani.shp      9-tehsil boundary
|-- dashboards\
|   |-- interactive_drought_dashboard.html
|   +-- drought_stage_timeline.html
|-- outputs\
|   |-- csv\          13 files     tables described in section 3
|   |-- figures\       8 PNGs      section 5
|   |-- maps\         22 files     section 6: 9 PNG + 9 PDF + contact + 2 md + prompt
|   |-- models\drought_rf_model.joblib
|   |-- rasters\      517 files    387 Sentinel-2 + 130 classified
|   +-- reports\      summary.md | summary.json | drought_dashboard_data.json
|                  workflow_report.pdf (this document)
|-- scripts\                       smoke_api.py, _render_maps.py, helpers
+-- src\
    |-- main.py                    9-stage orchestrator + verify_outputs
    |-- gee\      auth / download / soil_moisture
    |-- weather\  openmeteo.py     SPI computation
    |-- processing\  indices / classify / maps.py (publication maps + QC)
    |-- ml\       random_forest / sampling
    |-- viz\      charts.py        8 figures + 2 dashboards + orchestration
    +-- api\      endpoints.py     FastAPI
```

10. Rerun commands

```
# Full pipeline (cached, ~32 s) - never add --force (re-downloads 130 months)
python src\main.py --gee-project dark-influence-491307-n3

# API smoke test (9/9)
python scripts\smoke_api.py

# Maps + data audit + QC only (~40 s)
python scripts\_render_maps.py
```

11. Bottom line

End-to-end green: pipeline exit 0, 564 files / 0.79 GB, every verify stage passes except the 3 documented Sentinel-2 cloud gaps.

9 publication maps (PNG 3900 2550 px + vector PDF) with the QC report showing ALL PASS on all 7 checks.

data_audit.md validates area arithmetic (0.000% error), stage shares (100.000000%) and area-weighted means, while surfacing every data caveat instead of hiding it.

Random Forest baselines: 77.6% accuracy on composite-CDI stages and 90.7% on SPI-3-only stages.

No input data, CDI formula or stage threshold was changed only visualization, audit and verification code.